

Personal cybersecurity

Frameworks, habits, confidence

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Goals

Help you. . .

- ▶ Recognize and respond to cyber risk
 - ▶ Adopt simple, useful frameworks to manage risk
 - ▶ Aim to reduce risk, not eliminate it
- ▶ Adopt safe cyber habits
 - ▶ Plan and prepare
 - ▶ Focus on habits, not tools
- ▶ Be informed and confident in your cyber decisions
 - ▶ Have a reference list of trustworthy tools and information

Outline

1. Fundamentals
2. Your security
3. Toolkit overview
4. Conclusion

Fundamentals

Outline:

- ▶ Definitions
- ▶ Axioms
- ▶ Frameworks

Definitions

What is cybersecurity?

- ▶ Identify, assess, address, and monitor relevant cyber threats

Axioms

Outline:

1. The Internet is
2. Tradeoffs exist
3. Cyberattacks happen

The Internet

- ▶ The Internet is public
 - ▶ Anyone can use it
- ▶ The Internet is open
 - ▶ Anyone can see Internet traffic
- ▶ The Internet is forever
 - ▶ [Internet Archive](#) (Wayback Machine)

Tradeoffs

- ▶ Tradeoffs exist
 - ▶ Security versus cost
 - ▶ Security versus convenience
 - ▶ Trust versus time
- ▶ Tradeoffs change
 - ▶ Technologies change
 - ▶ Threats change
 - ▶ Values change
- ▶ Tradeoffs depend on context

Cyberattacks happen

- ▶ You can't control...
 - ▶ Whether cyberattacks happen (they will)
 - ▶ Whether you achieve perfect safety (you can't)
- ▶ You can control...
 - ▶ How informed you are
 - ▶ Know what to do and where to go
 - ▶ How prepared you are
 - ▶ What balance *you choose* among tradeoffs
 - ▶ What laws you support (privacy, accountability)
 - ▶ What grace you show yourself and others
 - ▶ More perfect, never perfect

Your turn

Which is riskier?

- (a) Your bff AirDrops a file to you irl
- (b) A stranger sends you a file over the Internet

Your turn

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Context:

- ▶ Your bff
 - ▶ Every file your bff has shared with you recently has been infected with malware
 - ▶ The file seems unusually large
- ▶ The stranger
 - ▶ The stranger is (allegedly) an employee at Microsoft
 - ▶ The file is a security update whose checksums agree with those on Microsoft's website

Frameworks

Outline:

- ▶ CIA triad
- ▶ Risk
- ▶ AAA
- ▶ OODA loop

CIA triad

- ▶ Confidentiality
 - ▶ Unauthorized entities can't access resources
 - ▶ Opposite: disclosure
- ▶ Integrity
 - ▶ Resources are reliable
 - ▶ Opposite: alteration
- ▶ Availability
 - ▶ Authorized entities can access resources
 - ▶ Opposite: denial of service

How to use it:

- ▶ To understand and assess security

Risk

- ▶ Asset
 - ▶ Something of value
 - ▶ Can be physical, digital, or intangible
- ▶ Vulnerability
 - ▶ A weakness
- ▶ Threat
 - ▶ A potential danger
- ▶ Risk
 - ▶ A potential loss that occurs when a threat exploits a vulnerability of an asset

How to use it:

- ▶ To understand and manage risk

AAA

- ▶ Identification
 - ▶ Who are you?
- ▶ Authentication
 - ▶ How do I know you are who you say you are?
- ▶ Authorization
 - ▶ What are you allowed to do?
- ▶ Accounting
 - ▶ What did you do?

How to use it:

- ▶ To understand interactions with service providers

Factors of authentication

- ▶ Something you know
 - ▶ Password
- ▶ Something you have
 - ▶ Cell phone
- ▶ Something you are
 - ▶ Face, fingerprint

How to use it:

- ▶ To understand MFA
 - ▶ Use two or more *different* factors during authentication

OODA loop

1. Observe
2. Orient
3. Decide
4. Act

How to use it:

- ▶ To make deliberate decisions under stress

Your security

Outline:

- ▶ Account access
- ▶ Software
- ▶ Devices
- ▶ Networks
 - ▶ Home router
 - ▶ Public Wi-Fi
- ▶ Relational
- ▶ In public
- ▶ Recovery

Account access: Your turn

- ▶ How many passwords do you have?
 - ▶ Give an order-of-magnitude estimate: 1, 10, 100, 1000,...
- ▶ How do you manage your passwords? (Select all that apply.)
 - (a) I use the same password or similar passwords for most or all of my accounts.
 - (b) I write down my most commonly used passwords on a piece of paper and keep it on or near my computer.
 - (c) I save them in my Web browser (for example, using the “remember me” feature).
 - (d) I write down my passwords in a notebook and look them up when I need them.
 - (e) I use a password manager.
- ▶ What are the tradeoffs for each method of password management listed above?

Account access: To do

- ▶ Use a password manager
- ▶ Review the security settings on your accounts
 - ▶ Password strength
 - ▶ MFA enabled?
 - ▶ Security alerts
 - ▶ Account-recovery options
- ▶ Implement MFA on high-value and high-risk accounts

Password best practices

- ▶ Don't reuse passwords
- ▶ Make passwords sufficiently long
- ▶ Don't use security questions
 - ▶ Why not? OSINT, social engineering

Password managers

Benefits:

- ▶ Generates and stores unique, complex passwords
- ▶ Integrates with Web browsers and mobile devices
- ▶ Auto-fills login credentials only on valid website

Risk:

- ▶ Keys to the kingdom
 - ▶ Make your master password strong

Password managers: How to start

1. Review
2. Choose
3. Install
 - ▶ App: on PC, phone
 - ▶ Web-browser plug-in: on PC
4. Enter credentials (for each account)
5. Update passwords
 - ▶ In the password manager *and* on the account
 - ▶ Use random-password generator feature

Software: To do

- ▶ Install software from trusted sources only
- ▶ Review the security settings and permissions of applications
- ▶ Uninstall unused software

Devices: To do

- ▶ Review the security settings on your digital devices
 - ▶ Login settings
 - ▶ Lock-screen timer
 - ▶ Encryption
- ▶ Align the security settings to the device
 - ▶ Tradeoff: Security versus cost and convenience
 - ▶ Considerations:
 - ▶ Device mobility
 - ▶ Data value
- ▶ Regularly apply software patches and updates
 - ▶ Automate?

Networks

Outline:

- ▶ Be wary of public Wi-Fi
- ▶ Harden your home router

Public Wi-Fi

Risks:

- ▶ Rogue access points
- ▶ Man-in-the-middle attacks

Habits:

- ▶ Enable MFA on sensitive accounts
- ▶ Avoid sensitive transactions
- ▶ Use HTTPS, not HTTP
- ▶ If HTTPS is not enough, then use a VPN

Home router: To do

- ▶ Change default credentials
- ▶ Enable strong encryption
 - ▶ Best: WPA3
 - ▶ Second best: WPA2
 - ▶ Don't use: WEP, WPA
- ▶ Enable automatic updates
- ▶ Disable risky features
 - ▶ Remote management
 - ▶ Wi-Fi Protected Setup (WPS)
 - ▶ Universal Plug and Play (UPnP)
- ▶ Create a guest network
 - ▶ Enable a separate, strong password

Relational

Outline:

- ▶ Social engineering
- ▶ Phishing

Social engineering

- ▶ An attempt to manipulate humans

Commonly used tactics:

- ▶ Scarcity
- ▶ Urgency
- ▶ Intimidation
- ▶ Authority
- ▶ Trust
 - ▶ Social proof
 - ▶ Familiarity
- ▶ Likeableness
 - ▶ Pity

Phishing

- ▶ Social engineering over communication technology

Variants:

- ▶ Phishing
 - ▶ Via e-mail
- ▶ Vishing
 - ▶ Via voice call
- ▶ Smishing
 - ▶ Via SMS

How to detect phishing

(Often, not always!)

- ▶ Is unexpected
- ▶ Is from an unusual source (e-mail address, phone number)
- ▶ Uses a generic greeting
- ▶ Uses social-engineering tactics

Social engineering: To do

- ▶ Listen to your Spidey sense
- ▶ Slow down and question
- ▶ Verify by using another route
- ▶ If it's a phish, then report and delete

Your turn

You receive the following e-mail from your bank:

*We have detected suspicious activity on your account.
Please use the following link to verify your identity and
account activity within 24 hours or your account will be
frozen: [verify account](#)*

You use this bank account to pay multiple important bills, several of which come due in the next few days.

What do you do?

In public: To do

- ▶ Configure secure device settings in advance
- ▶ Don't leave your devices unattended
- ▶ Beware shoulder surfing
 - ▶ Bystander, binoculars, camera
- ▶ Beware tailgating and piggybacking

Recovery

A cyber incident happened!

- ▶ Forgotten password
- ▶ Account compromise
- ▶ Hardware failure
- ▶ Ransomware

Now what?

- ▶ Account resets
- ▶ Data backups

Account resets

- ▶ Forgotten password
 - ▶ Receive e-mail with temporary code or link
 - ▶ Enter recovery code (issued at account creation)
- ▶ Lost MFA device
 - ▶ Install authenticator app on second device
 - ▶ Capture and securely store static setup key
- ▶ Locked account
 - ▶ Wait
 - ▶ Request automated password reset
 - ▶ Contact customer support

Data backups

- ▶ Adopt a realistic backup routine
- ▶ Protects against accidental file changes, hardware failure, ransomware
- ▶ Options:
 - ▶ External hard drive
 - ▶ Cloud storage
- ▶ Considerations:
 - ▶ Privacy
 - ▶ Availability
 - ▶ Criticality of data
 - ▶ Urgency of access

Toolkit overview

- ▶ Password manager
- ▶ Anti-malware software
- ▶ Data backup
- ▶ Check suspicious file or URL: [VirusTotal](#)
- ▶ Check data breach: [HaveIBeenPwned](#)
- ▶ Report phishing: [APWG](#)

Caution:

Don't upload sensitive information

Conclusion: What we aimed for (goals)

Help you. . .

- ▶ Recognize and respond to cyber risk
 - ▶ Adopt simple, useful frameworks to manage risk
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- ▶ Adopt safe cyber habits
 - ▶ Plan and prepare
 - ▶ Focus on habits, not tools
- ▶ Be informed and confident in your cyber decisions
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Conclusion: What we learned

- ▶ Realities
 - ▶ The Internet is public, open, and forever
 - ▶ Tradeoffs exist
 - ▶ Cyberattacks happen
- ▶ Frameworks
 - ▶ CIA triad
 - ▶ Risk
 - ▶ AAA
 - ▶ OODA loop
- ▶ Habits
 - ▶ Prepare in advance
 - ▶ Be alert for social-engineering tactics
 - ▶ Slow down, question, and verify
 - ▶ Make friends with a few security tools
 - ▶ Show grace to yourself and to others

Conclusion: What we'll do

- ▶ Today
 - ▶ Share what you learned with a friend
- ▶ This week
 - ▶ Enable MFA on sensitive accounts
- ▶ This month
 - ▶ Start using a password manager
 - ▶ Adopt regular routines to...
 - ▶ Update software
 - ▶ Back up data
- ▶ This year
 - ▶ Configure secure settings for accounts and devices

Thank you

- ▶ Presentation by: Stephen Wolff
- ▶ Presentation available at:
<https://shw3512.github.io/teach/cysec-g30/>
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